



DEPARTMENT OF ICT AND NATURAL SCIENCE

NORDARK WP5

NORDARK Digital Twin (DT)

User Manual

Author:

Evi Elisa Ambarita, Arne Styve

Version:

1.0

26th May 2025

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Revisions

Version	Date	Author(s)	Comment
1.0	25. May 2025	Evi Elisa Ambarita and Arne Styve	Initial version

1 Introduction

Nordark WP5 is digital twin-based simulation and visual analytics to assess outdoor conditions. The Nordark digital twin (DT) is a tool that support the design, planning, and assessment of suitable outdoor lighting environments, as well as the specification and implementation of visual analytics tools to support data analysis.

DT will be used for designing operations in the two study sides (pre-intervention phase) as well as supporting the analysis of results related to post-interventions.

- Pre-intervention phase: used to define the proper location and operation procedures related to both envisioned lighting solutions (WP4) as well as traps and acoustic sensors (WP3).
- Post-intervention analysis: used to visualize psychological (WP1) and physiological (WP2) data collected from the different study sites. This analysis involves identifying patterns of interest, trends, and correlations with other variables, such as environment and climate data.

The process of designing DT for intervention planning and post-intervention data analysis is illustrated in Figure 1. The digital twin is developed using Unity with the High-Definition Rendering Pipeline (HDRP). Several Unity assets are used, the main ones being:

- The Mapbox Unity SDK, to display a map of any location.
- Vegetation Studio Pro, to create vegetation randomly on user-defined areas.
- Photorealistic lights (IES), to apply IES files to lights (more on this later).
- A collection of vegetation assets

Detailed informartion regarding the technical documentation is provided in "Technical document-ation.docsx". Additionally, the work related to the development of the digital twin is publicly available in the form of research articles, as listed in the Bibliography.

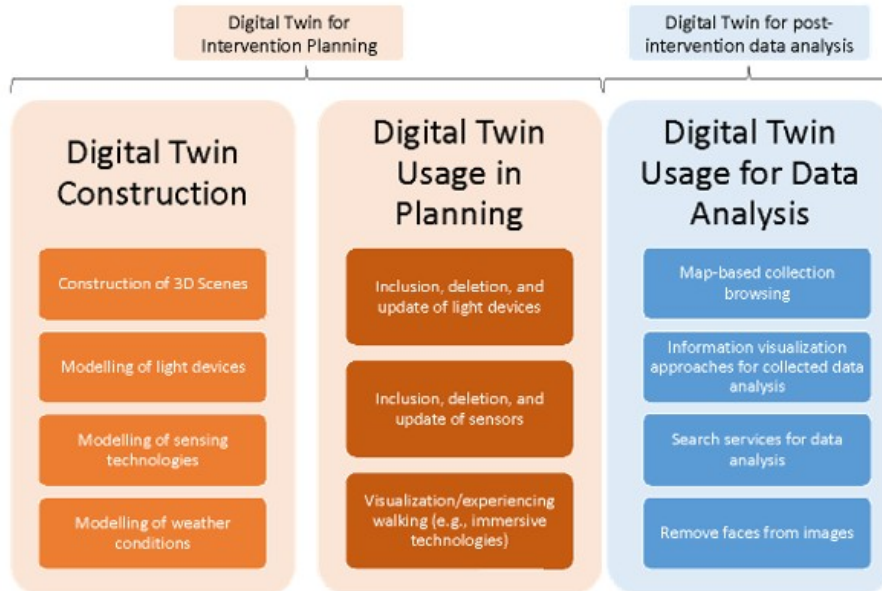


Figure 1: The illustration of DT proces from requirement from features

2 Getting Started

Before we start to install the software, make sure you have a compatible PC. As the PC we are currently using to run the digital twin have the following specification (purchased juni 27, 2022):

- Graphics card: Nvidia GeForce RTX 3080 10GB GDDR6X
- Harddisk: 1 1TB M.2 PCIe NVMe SSD (Boot) + 1TB 7200RPM SATA 6GB/s (Storage)
- Memory (RAM): 32GB DDR4 3200MHz
- Processor (CPU): 1 AMD Ryzen(TM) 9 5950X (16-Core, 72MB Total Cache, Max Boost Clock of 4.9GHz)

So if you can find something close (focus mostly on graphics card and memory) it should be fine.

Step 1a. If you have an access to the NORDARK github, go to release page by entering the following URL <https://github.com/NTNU-IE-IIR/NORDARK/releases> in your browser. Download the *Build-Standalone.zip* file for the lastest version (v0.12.2), selecting the appropriate version for your operating system. We provide builds for Linux64, macOS, and Windows64.

Step 1b. If you do not have access to the NORDARK github, get the *Build-Standalone.zip* file corresponding to your operating system.

Step 2. Extract the downloaded archive.

Step 3. Run the software by double-clicking on the *NORDARK-WP5.exe* file.

Step 4 Since this software is build in Unity, a small screen adjustment may be required to ensure all menus are visible. Try mazimizing and minimizing the software window while displaying the *Light Computation* feature. Once you see the *Split screen* option at the buttom of *Configurations* column, the screen size is properly set for full functionality.

Enjoy exploring!

2.1 Mouse and Keyboard Controls

Some information on this section are also provided on the software: *Help*, then *Display controls*

Hold right click mouse for rotating the camera

Scrolling mouse or W and S or up-and-down arrow for zooming in and out

A and D or left-and-right arrow for moving left and right

Q and E for moving up and down

2.2 The required files

To follow along with the examples in this user manual, you will need the following files:

- Path1mDATAStudyCase2WGS84.geojson – used for data visualization feature in section 3.7, representing the walkability path in Lerstad, Ålesund.
- case3.csv – used as an example of CSV file to be converted to geojson format, for section 4.3.
- convert_CSV_to_geojson.py – a python script used to convert the CSV file to geojson format, for section 4.3.

-
- EnvironmentalPsychology.geojson - used for representing environmental psychology along the selected path, coming from Work Package 1 in section 5.1,
 - case3.geojson – used for comparing the measured data with the simulated data. The data came from Work Package 4, visualized in section 5.2 for Scenario 1 100% (November - no snow) in Sävja, Uppsala - evening measurements - ground level (h: 0.00 m).
 - case55.geojson – also used for comparing the measured data with the simulated data, in section 5.2. Scenario 1 100% (January - snow) in Sävja, Uppsala - evening measurements - ground level (h: 0.00 m).
 - Arne_questionnaire_opinion.geojson – used for visualizing Arne’s questionnaire. The data came from Work Package 6, as detailed in section 5.3
 - Arne_questionnaire_5var.geojson – also used for Arne’s questionnaire for viewing residents’ responses to each individual question. The data came from Work Package 6, as detailed in section 5.3.
 - lifestyle_wildlife.geojson – used for visualizing data in exploring residents’ views on the balance between a comfortable urban lifestyle and wildlife protection. The data came from Work Package 6, as detailed in section 5.4.

3 Features and Functionality

3.1 Scene

3.1.1 Predefined Scenes/locations

This feature provides two predefined locations: Ålesund and Uppsala. The Ålesund location covers the Lerstad area in Norway, while the Uppsala location corresponds to Sävja area in Sweden.

You have the option to create a new location, by inputting the latitude and longitude of your chosen site, which you can find from Google Maps. Figure 2 provides an example with coordinates corresponding to NTNU in Ålesund. With camera adjustment, you will see the location on the right side of Figure 2.

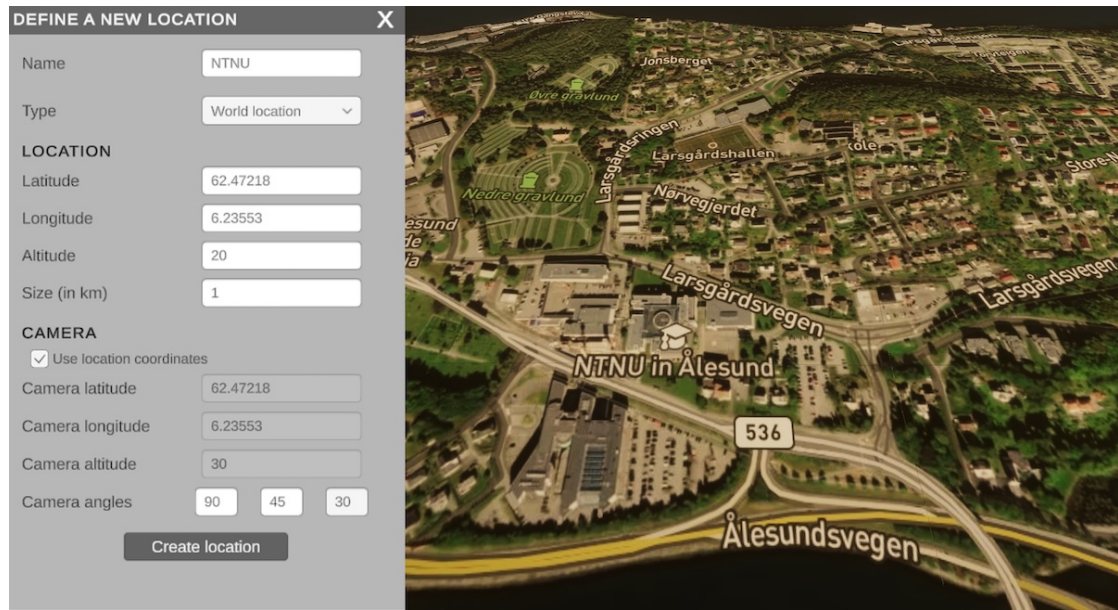


Figure 2: An example of creating a new location. Left: the required information to define a new location. Right: the created location with some adjustment on the camera position

This feature also offers 15 options for customizing the ground appearance, including streets, outdoors, dark, light, satellite, satellite street, commercial, farmland, forest, grass, path, residential, roads, water, and wetland. You can choose to include or exclude the buildings by checking the box of *Display buildings*. All these options are illustrated in Figure 3. Note that the ground customization is only applicable to created locations and does not affect the Ålesund and Uppsala locations. Moreover, you have the option to delete the selected location.

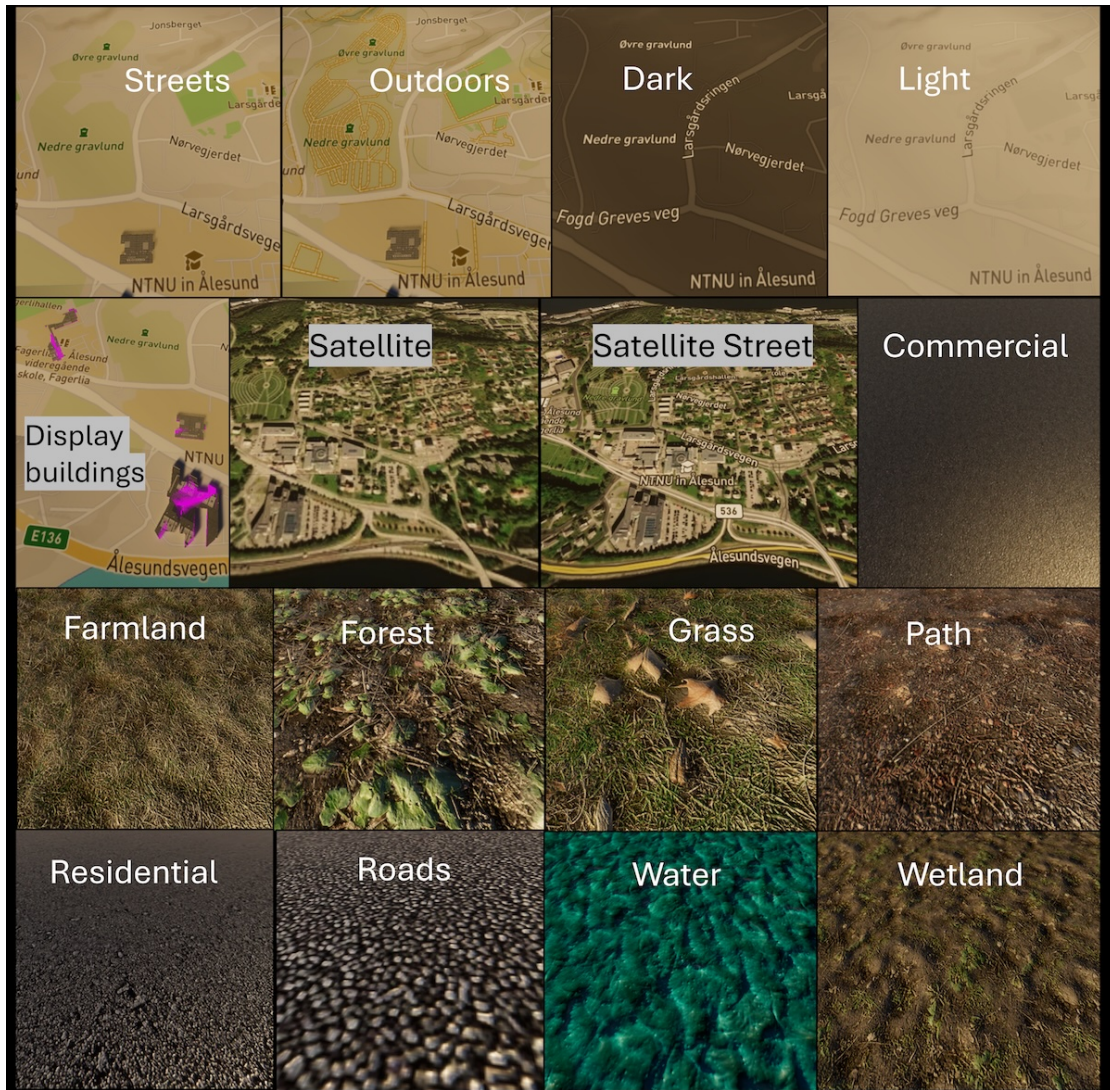


Figure 3: Ground option

Since the sun and moon positions change over time, you can set the desired time by adjusting the hour, date, month, and year. Figure 4 provides an example showing how the sun's position varies depending on the month. In this example, the Ålesund location is shown at the same hour and date for each month.



Figure 4: Sky option

In the *Weather* column, the Scene feature provides five cloud types: clear, sparse, cloudy, overcast, and stormy. You can also customize weather conditions, such as rain, snow, and fog, with intensity level ranging from 0 to 1.

3.2 Vegetation

Vegetation feature covers the vegetation options, such as *Biome areas* and *Vegetation objects*. The *Display nodes* is to show the ...

You can customize the forest density by adjusting the density bar ranging from 0 to 1. Figure 5 shows the appearance of the forest in Uppsala with different density settings.



Figure 5: Density scale in Uppsala

In the *Vegetation objects*, you can create a custom area with six different types of 3D vegetation models. To add a tree, click the *add* button or press the "I" keyboard, then left-click to place the tree. You can adjust the tree's rotation and change its vegetation type. To move a tree, click the *Move* button or press "M". To delete a selected tree, click *Delete* button or press the "Delete" key. Once you have finalized the position, type, and rotation, press "Esc" to deselect the tree. To select a different tree, click on the trunk rather than the leaves. If you want to select multiple trees, hold "Ctrl" while clicking additional ones.

3.3 Lights

Light feature covers illumination from light poles. Several pre-made light poles are available, or you can create your own. To select an existing light pole, you can either (i) click on it directly, or (ii) click the *select* button then double-click the desired pole. The selected pole(s) is marked with a pyramid on the pole, as shown in Figure 6.

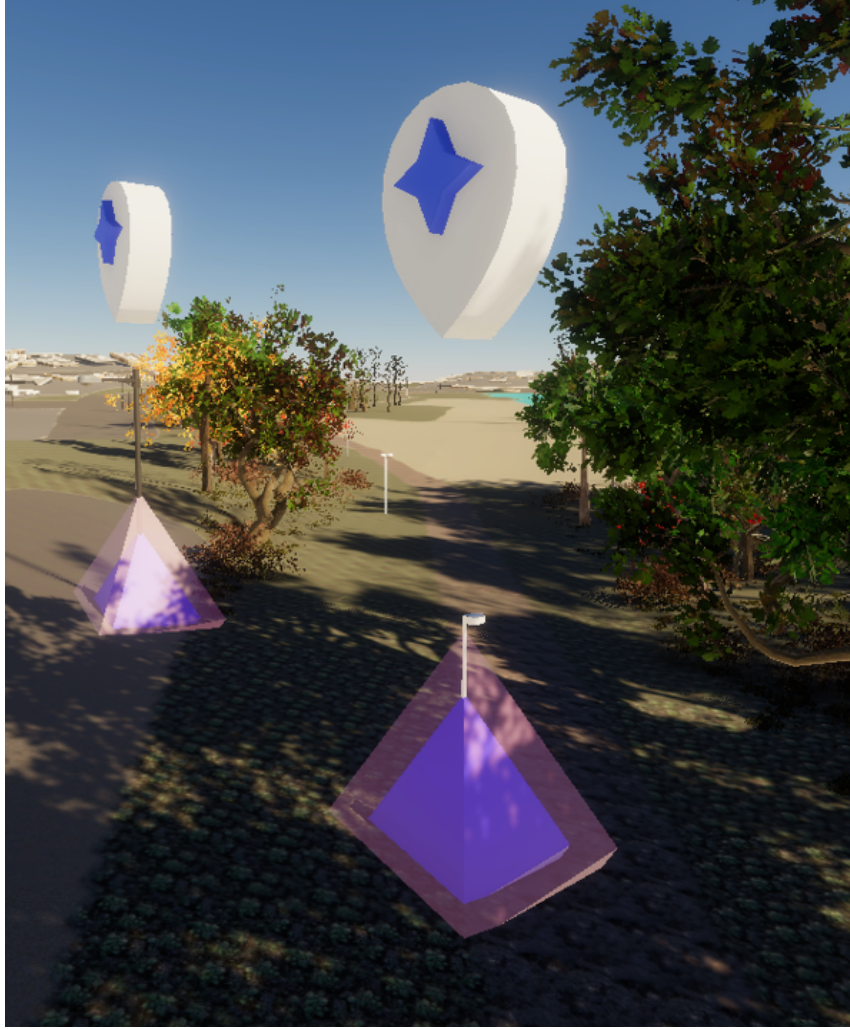


Figure 6: Marking the selected poles

To add more light poles on the selected scene, use the *add* buttons or press "I" keyboard, then left-click to replace the light pole. You can customize the height and rotation of the selected pole. To move the selected pole, click the *Move* button or press "M". To delete the selected pole, click *Delete* button or press the "Delete" key. Once you have finalized the position, height, rotation, and the model, press "Esc" to deselect the pole. If you want to select multiple poles, hold "Ctrl" while clicking additional ones.

This feature provides 6 different types of 3D-modelled lights, as shown in Figure 7. The top section shows the light during the day, while the bottom section shows them at night. We provide four types of IES file. IES file defines how light from a lamp is distributed in a space. This data is provided by many manufacturers so that lighting designers can realistically simulate how a project will look when a specific light source is used Library 2024. You also have the option to upload your own IES file to customize the lighting.

The *Highlight light poles* checkbox highlights all available light poles. However, to view the result properly, you need to zoom the camera out; otherwise, only a yellow glow will be visible. The *Display light poles* checkbox shows all available light poles, including both default and customized ones.

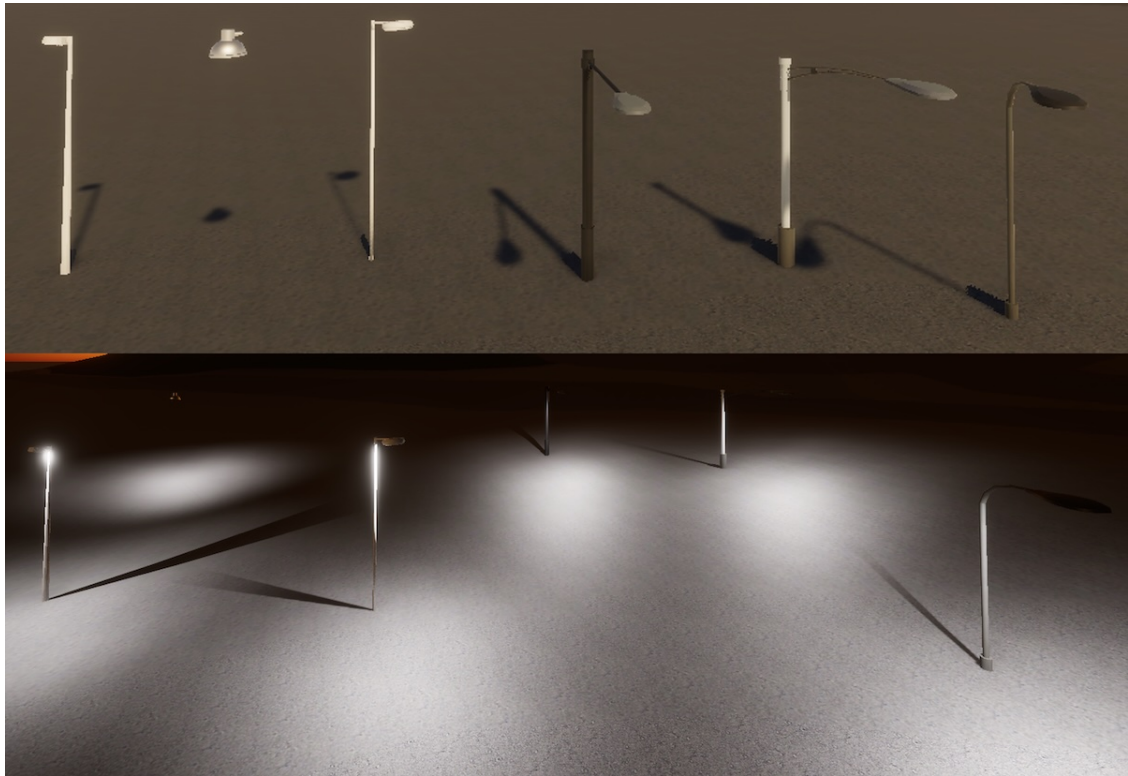


Figure 7: 3D model of light poles. Upper: on the day light. Bottom: on the night

3.4 Cameras

Camera feature allows you to manage camera settings, including properties such as sensor size, ISO, shutter speed, lens length, and aperture. You can also add, remove, and modify cameras. To add a camera, click the *Add a new camera* button and specify the camera properties. Note that the most recent settings will be automatically saved to the selected camera. The *Update the current camera position* button adjusts the view in the *Camera preview* to match your screen's view. On the other hand, the *Move to the current camera position* button changes the view on your screen to match the one in the *Camera preview*. You can also delete the selected camera if needed.

3.5 Street View

The *street view* feature allows users to walk in the scene as if person would do. Follow these steps to use it: (1) adjust the camera to frame the desired scene, (2) press the "Esc" key to release the camera, (3) select a character from the three available options: insect, child, or adult, with heights ranging from 2 cm to 188 cm. To readjust the camera, press "Esc" again. Note that in this feature, the "Q" and "E" keys, using to move up and down, do not work.

3.6 Light Computation

This feature provides the luminance computation. The luminance is a photometric measure of the luminous intensity per unit area of light traveling in a given direction. It is expressed in candela per square meter (cd/m²). In Unity, the luminance is computed from the pixel color and the camera exposure. Three methods are available for visualizing luminance: a luminance map, a luminance graph along a line, and a luminance heatmap in a rectangle.

To visualize the luminance on the map, explore *luminance map* column and check the *Display*

luminance map box. You also have the option to include or exclude the *Log scale*, as shown in Figure 8. The luminance map will update based on the selected time and season. To check the luminance at a specific point, hover your cursor over the desired location. The luminance value for that point will appear in the *Luminance Map* column under the *Log scale* checkbox.

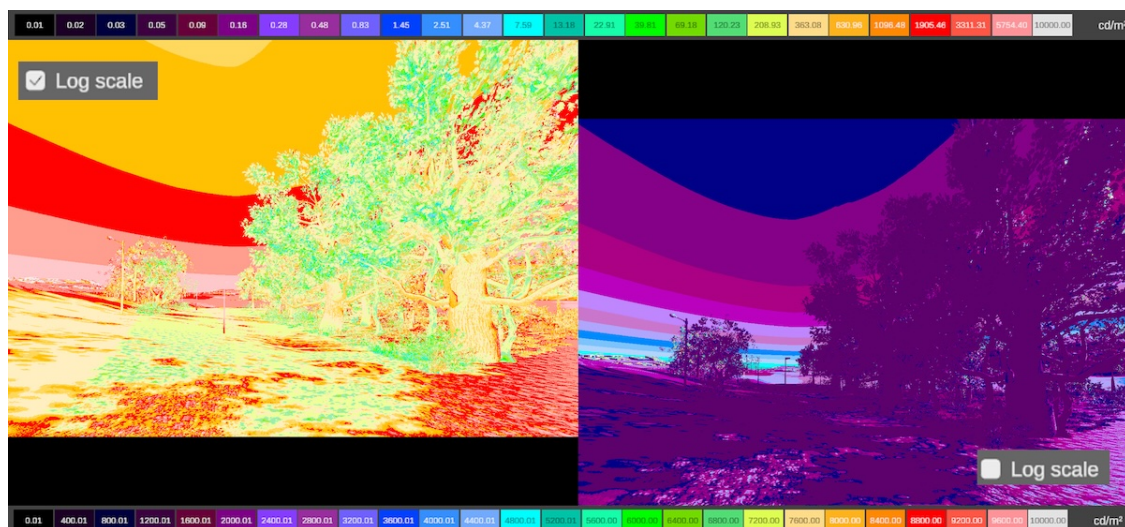


Figure 8: Luminance map on Ålesund with log scale (left) and without log scale (right)

To visualize the luminance in a graph, explore the *Line Visualization* column. First, you need to create a line using one of the two methods:

- *Create measure line manually*: click the button to start drawing the line. Place the cursor where you want the line to start, left-click for the starting point, and right-click for the ending point. To create a connected line with multiple segments, continue left-clicking until you're done, then right-click to release the line drawing.
- *Create measure line around light pole*: click the button and select a light pole. You'll need to input three variables for the line: length, angle relative to the light pole, and distance between measurement points. For example, if you want the line to be 10 meters long and take measurements every meter, set the distance to 1 meter.

Next, check the *Display graph* box to show the graph along the drawn line. Figure 9 provides an example with the blue circle showing two options for visualizing luminance: column graph and line graph.



Figure 9: One example of visualizing the luminance in line graph

To visualize the luminance as a heatmap, explore the *Grid Visualization* column. First, you create a rectangle by clicking the *Create measure rectangle* button, then position the cursor of where you want the rectangle. Start the drawing by left-clicking, adjust the size as needed, then release the drawing with a right-click. Next, check the *Display heatmap* box. If you want to view the location of a certain heatmap, bring the cursor on the heatmap. A red line will appear within the rectangle, indicating the location. Figure 10 provides an example, showing the cursor on the heatmap and the red line in the rectangle, captured with a phone camera. You can also include or exclude the values on the heatmap using the checkbox *Display values on heatmap*.

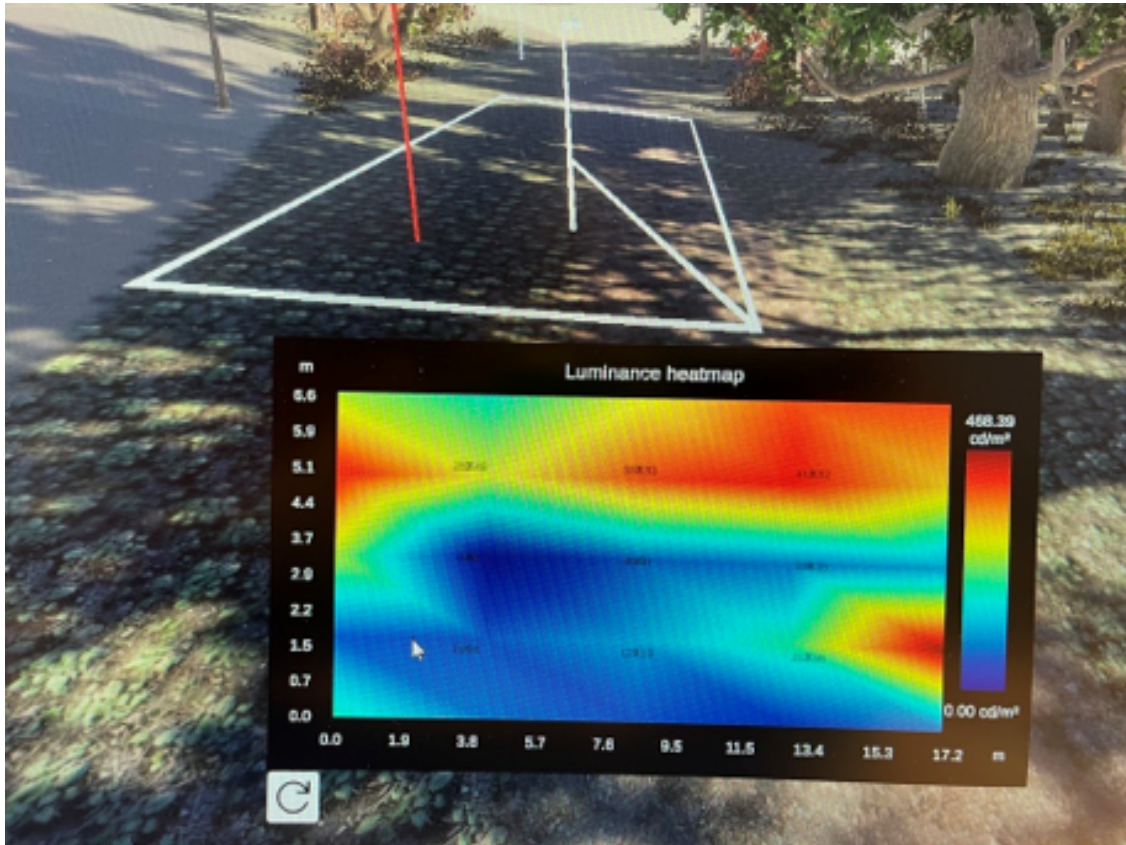


Figure 10: One example of luminance heatmap, showing the cursor on the heatmap and a red line on the rectangle

You can save the luminance results (graph and heatmap) by exporting them as GeoJSON and CSV files. For graph visualization, you can later import the saved results to compare them with new ones.

In the *Configurations* column, you can split the screen into up to four sections. This is useful for comparing luminance maps across different settings. An example of this can be seen in Figure 11, where the reference pole is highlighted with a yellow circle. The top-left section shows the default setup with only two light poles. The top-right section has no light poles in the area. The bottom-left section includes four light poles, three of which are positioned toward the left side. The bottom-right section also shows four light poles, but three are positioned toward the right side. The image was captured during the same setup at midnight.

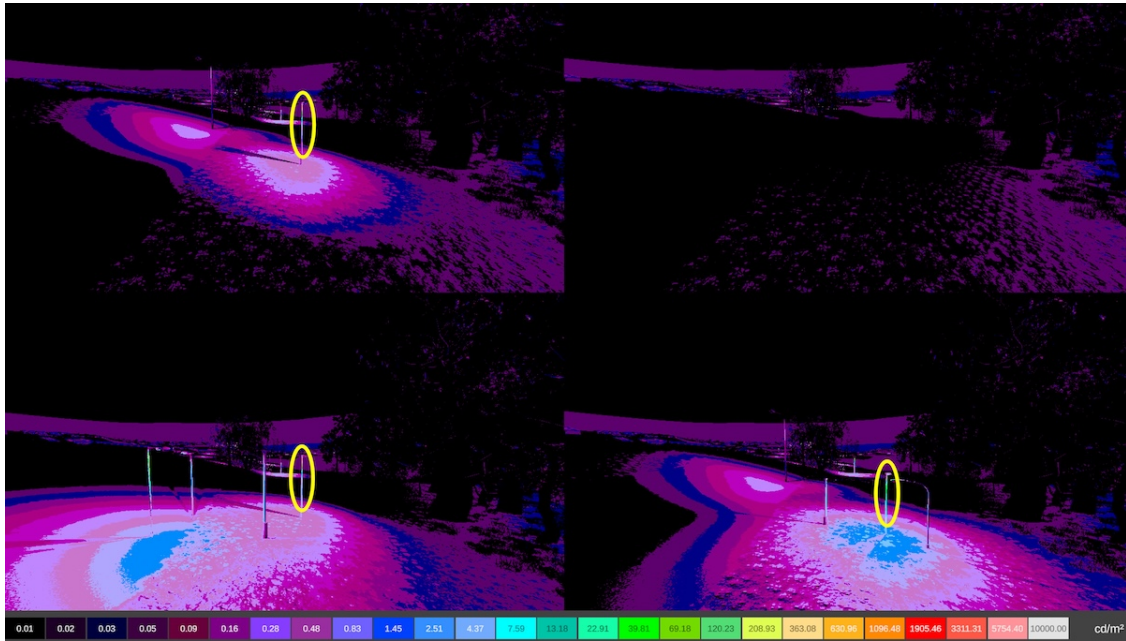


Figure 11: Luminance map comparison for four configurations

3.7 Data Visualization

This feature supports visualization exclusively for data in the GeoJSON format. This format requires a geometry property that defines the data's location. The geometry can be represented as either a Point or a LineString, both using latitude and longitude coordinates. These coordinates can be obtained via Google Maps or by following the steps outlined in Section 4.3. The feature visualizes data by scaling values between 0 and 1, displayed in a heatmap and a bar chart. In this context, 0 (shown in red) represents the lowest or worst value, while 1 (shown in blue) indicates the highest or best. Therefore, the data you want to visualize must be quantitatively scalable. If your dataset consists of qualitative descriptions that cannot be translated into a numerical value, it cannot be visualized using this feature. An example of compatible data is provided, which is the walkability analysis of different paths in the Ålesund area.

First, you need to download a file named "Path1mDATASTudyCase2WGS84.geojson". Then, on the *Data visualization* feature, click the *Add dataset* button, and select the file. With screen adjustments, the final view appears as shown in Figure 12. At the bottom, you see a heatmap bar ranged from red (0) to blue (1). This heatmap indicates the walkability in different path based on the selected scenario. The red areas represent the most challenging sections, while orange, yellow, green, and blue indicate progressively easier terrain.

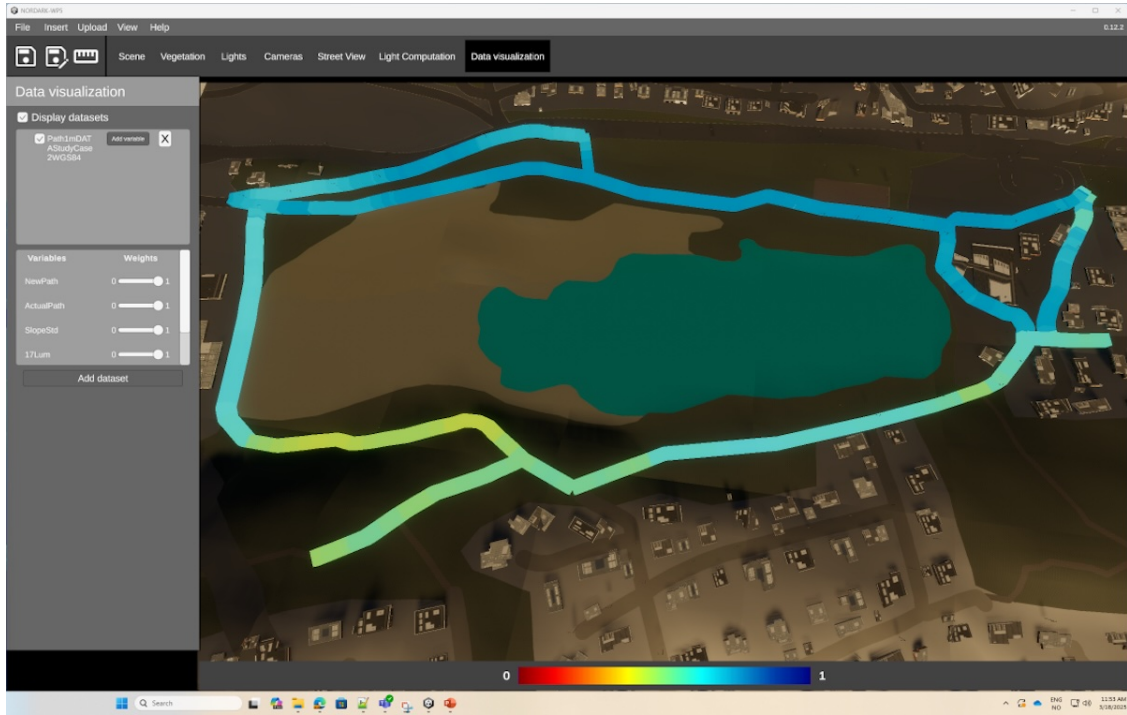


Figure 12: An example of data visualization

On the left side of Figure 12, you will find several variables with adjustable weights ranging from 0 to 1, including *NewPath*, *ActualPath*, *SlopeStd*, *17Lum*, *5Lum*, and *ActualLum*. Modifying these variables allows you to customize the desired scenario. To better understand these variables, we provide Figure 13.

- (a). The difficulty level is assessed based on the actual path, reflecting real ground conditions.
- (b). The assessment focuses solely on the slope condition.
- (c). The difficulty level is evaluated considering the actual lighting conditions.
- (d). The analysis incorporates three variables simultaneously: actual path, slope condition, and actual lighting.
- (e). While considering the actual path and slope condition, the light condition is adjusted to 17 and 5 luminance
- (f). The difficulty level is assessed using the slope condition in combination with the new path. The new path includes modifications to both ground and lighting conditions, aiming to make the entire area more walkable.

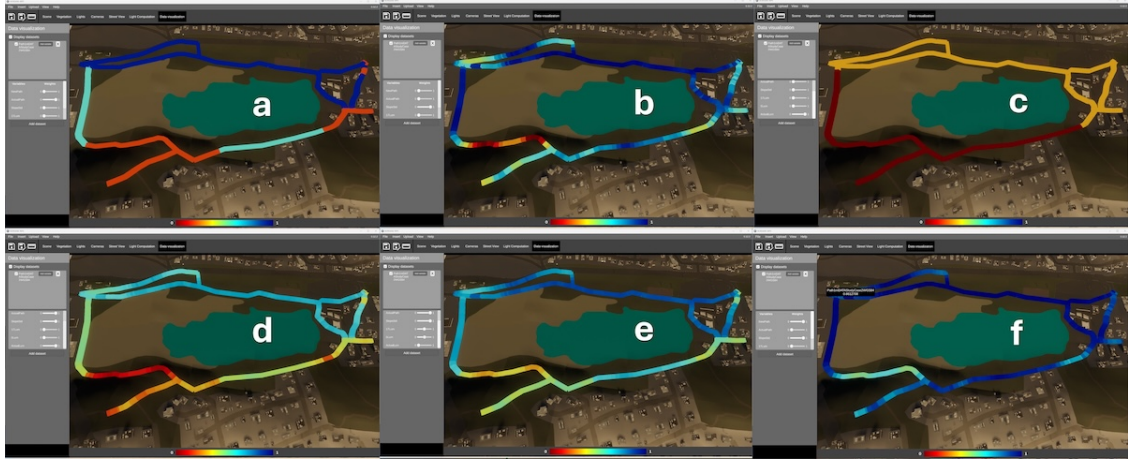


Figure 13: Data visualization on the walkability based on the considered variables. (a) considering the actual path, (b) considering the slope condition, (c) considering the actual lighting condition, (d) considering the actual path, the slope condition, and the actual lighting, (e) considering the actual path and slope condition, while adjusting the light condition with 17 and 5 luminance, (f) considering the slope condition with the new path.

4 Task

4.1 Understanding the light design problem

Step 1. Setting the scenario

- In the *Scene* tab, set the location to Ålesund. For this task, consider the path marked by the red line in the lower-left corner of Figure 14.

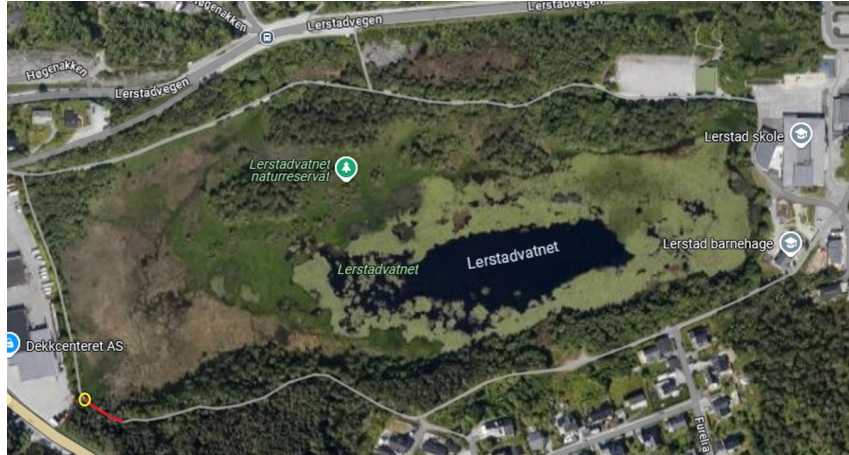


Figure 14: The analyzed path in Ålesund, the path marked by the red line in the lower-left corner

- Adjust the settings as follows: date to 25-December-2024, time to 16:30, clouds to *sparse*, weather to 0.5 *snow* and 0,5 *fog*. Figure 15 shows a view from the path shown in the yellow circle of Figure 14.

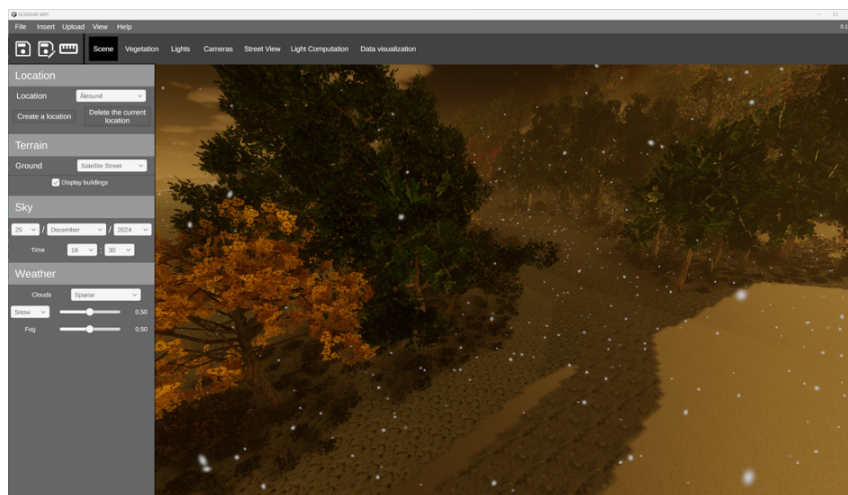


Figure 15: The view of the analyzed path

Step 2. Assessing the luminance condition

- By default, no light poles are designed on this path, as shown in the luminance graph in Figure 16.

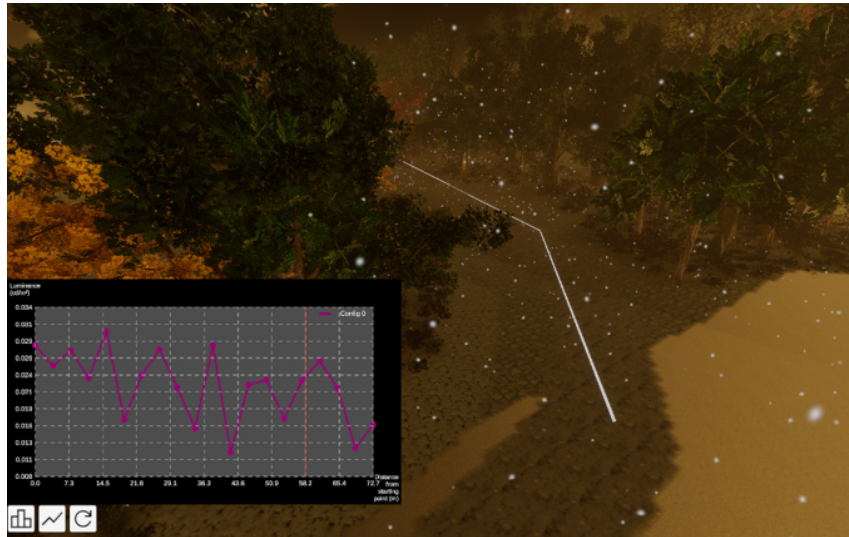


Figure 16: The luminance graph of the analyzed path, without light pole

- To generate this luminance graph, go to the *Light Computation* tab, draw a line along the path by clicking on *Create measure line manually*. Use left-click to start a new line and the right-click to finish the line. Then, check *Display Graph* box to visualize the luminance graph along the drawn line. To save the results, export them by clicking *Export (GeoJSON)* or *Export (CSV)*, name the file (for example: result1). To save the current situation, go to file, and click save, name the file (for example: test1)

4.2 Designing light interventions

Using the same setup from the previous task in 4.1 – where no light poles were present on the path – this task focuses on designing light poles along the path and comparing the results. By analyzing luminance across different setups, you can develop an optimized lighting model using Nordark digital twin. Various what-if scenarios can be explored in this digital twin before proceeding with installation.

Step 1. Creating light poles on the path

- In the *Lights* tab, click *Add*, then left click to place the pole. Adjust its height and rotation, choose the type and IES file.
- If you feel that there is not enough light poles on the path, add new ones by pressing "I" or clicking *add* in the *Lights* tab. An example is made with six light poles on the path, as seen in Figure 17.



Figure 17: An example of six designed light poles

Step 2a. Comparing luminance graphs

- Go to *Light Computation* tab, check the *Display graph* box, then click the refresh button below the graph.
- To compare luminance graphs, click *Import results* within the *Line Visualization* column, then select the file "result1". Two luminance graphs can be seen in Figure 18: the red line represents the new design with six light poles, while the green line represents the path without light pole (the imported one).

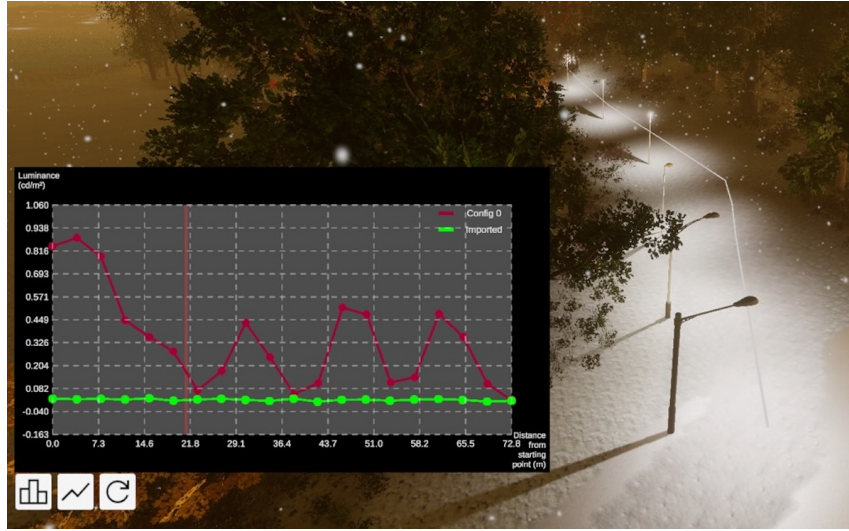


Figure 18: Comparison on luminance graphs: the red line represents the new design with six light poles, while the green line represents the path without light pole.

Step 2b. Comparing the luminance maps

- Go to *Light Computation* tab, on the *Configurations* column, set *number of contributions* to 2.
- On the *Scene config*, click the *Browse* button, then select the saved file "test1", then check the *Split screen* box.
- Check the *Display luminance map* within the *Luminance Map* column. The comparison result are shown in Figure 19.

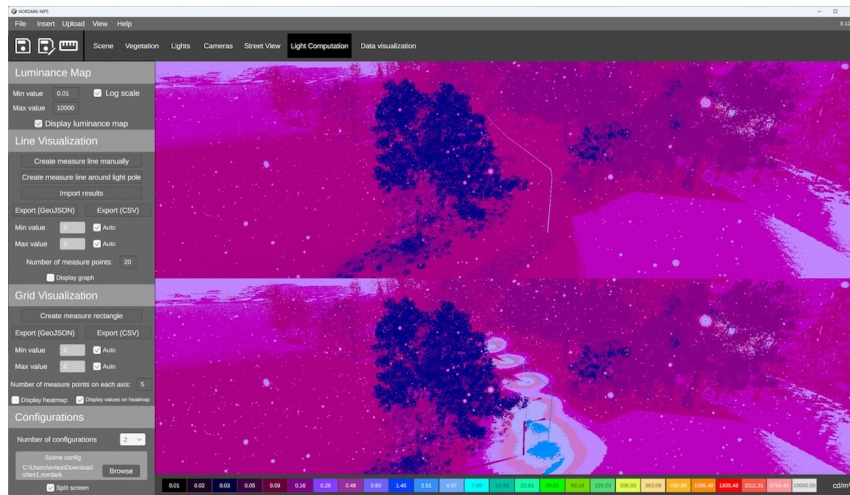


Figure 19: Comparison on luminance maps: Top section represents the path without light pole, while bottom section represents the new design with six light poles.

4.3 Converting data from CSV format to Geojson for Light Computation

The Nordark digital twin only supports geojson format to be imported, therefore, the original data needs to be converted to the geojson file. Given that GeoJSON is a format for encoding a variety of geographic data structures, the original file requires information regarding location. It can be

as a Point or a Linestring with latitude and longitude. There are two ways to find latitude and longitude information: (i) using Google Maps, and (ii) through the digital twin. For example, to find the coordinates of the red line shown in Figure 15, open Google Maps and locate the exact spot. Then, right-click on the location; the first row that appears will display a pair of numbers in the format [latitude, longitude]. Repeat this process to mark multiple points along the red line in Figure 15.

The other way to find latitude and longitude information is by using the digital twin through the *Light Computation* feature. First, draw a line along the path you want to analyze (see Section 3.6 for how to draw a line). Then, check the *Display Graph* box to show the luminance graph along that line. Click *Export (CSV)* to save the data. When you open the CSV file, you will see columns for latitude, longitude, altitude, and distance from the starting point (in meters). This lets you get the coordinates for any location where you draw the line.

When you look at the saved CSV file, you can see the format needed for the digital twin to read the data. The file includes the following columns: latitude, longitude, altitude (m), distance from origin (m), luminance (cd/m²), ground (Path/Forest), and config. In the GeoJSON format, latitude, longitude, and altitude (m) are grouped under a feature called geometry. The other values are listed as properties that belong to that geometry.

Through the observation on the saved CSV file, you see the information format that is required for the digital twin to be able to read. The format is the column list, namely latitude, longitude, altitude (m), distance from origin (m), luminance (cd/m²), ground (Path/Forest), and config. On geojson format, the latitude, longitude, and altitude (m) are represented as a feature called "geometry". While the rest are properties that represent the information belonging to a geometry.

To compare how the same data looks in both CSV and GeoJSON formats, use the *Light Computation* feature. Draw a line, display the graph, then export the results in both CSV and GeoJSON formats. Open both files to compare them. You will see how the columns in the CSV file match the structure of the GeoJSON file, including the feature collection, features, and properties. By understanding the similarities and differences between the two formats, you will be able to format your own Excel or CSV file correctly. Once your CSV file includes all the required columns, you can use our Python script below to convert it into a GeoJSON file. This python script is also provided as a file named "convert_CSV_to_Geojson.py".

```
import pandas as pd
import json
import os

# File paths
csv_path = r'D:\Project\case3.csv' # Change this path based on where you saved the CSV file
geojson_path = r'D:\Project\case3.geojson' # Change this name for a different output file

# Read the CSV file
df = pd.read_csv(csv_path)

# Build GeoJSON structure
geojson = {
    "type": "FeatureCollection",
    "features": []
}

for _, row in df.iterrows():
    feature = {
        "type": "Feature",
        "geometry": {
            "type": "Point",
            "coordinates": [
                row['longitude'],
                row['latitude'],
                row['altitude (m)']
            ]
        },
        "properties": {
            "luminance": row['luminance (cd/m2)'],
            "ground": row['ground'],
            "distanceFromOrigin": row['distanceFromOrigin (m)'],
            "config": row['config']
        }
    }
    geojson["features"].append(feature)
```

```
# Write to GeoJSON file
with open(geojson_path, 'w') as f:
    json.dump(geojson, f, indent=2)
```

Listing 1: Python script to convert CSV file to geojson format

An example CSV file is provided for the case described in Section 5.2. This file, named "case3.csv", contains light measurements from light poles PR1 to PR8. To convert this CSV file, enter the file name and path into the `csv_path` in the Python script.

The script uses the pandas library. If you haven't installed Python yet, please do so first. Then, open the Command Prompt and run the following command to install pandas: `pip install pandas`. Next, navigate to the folder where your script is located. For example: `cd C:\Desktop`

Project. Then run the script with: `python convert_csv_to_geojson.py`. Here are some key points about how the script works:

- Ensure the CSV file is saved and closed before running the script.
- The script expects the column names to match exactly (including parentheses and case).
- The coordinates are formatted as [longitude, latitude, altitude], which is standard for GeoJSON.
- The output GeoJSON will have each row as a Point feature with all the specified properties.
- To use a different name, simply modify the `geojson_path` variable.

4.4 Formatting data to Geojson for the data visualization

As mentioned earlier, the GeoJSON format requires geometry information regarding the latitude and longitude. You can get these coordinates by following the steps in Section 4.3. Each geometry can represent a different category you want to visualize. For example: in Section 5.1, each geometry represents one of three lighting conditions. In Section 5.3, geometries are used in two ways: (i) to represent different questions, and (ii) to represent different answers. How you define each geometry depends on the purpose of your visualization.

As explained in Section 3.7, the data visualization feature works by scaling values between 0 (worst) and 1 (best). So, before visualizing your data, you need to scale your values to fit within this range. Then, list your variables and include them as properties in the GeoJSON file.

An example of this process is shown in Section 5.1, using a task from WP1. Take a look at the original file WP1-Ålesund data_Day-Head torches.xlsx and the converted file EnvironmentalPsychology.geojson. In the original file, the **mean** values are rated from 1 to 5. These must first be scaled to fall between 0 and 1. In the converted GeoJSON file, you'll see three geometries—each one represents a column from the Excel file: daylight, head torch regular, and head torch customized. Each row from the Excel file becomes a set of properties under each geometry in the GeoJSON file. By reviewing both files and understanding how the conversion is done, you can apply the same approach to your own data.

You can also use the Python script mentioned earlier, with a few additional steps beforehand. First, you need to reformat your original file into a CSV format—this may involve switching the column headers to rows, or vice versa, depending on your data structure. Next, update the Python script by replacing the list of properties with your own variables. Also, make sure to double-check that the file paths in the script are correct.

5 Data from Nordark Work Packages

This section holds information about data collected from the other workpackages in the Nordark project. The type of collected data/information is discussed against the possibility to visualize the data/information in the current version of the Digital Twin, as well as provides suggestions on possible future ways of using the DT for visualization.

5.1 WP1: Analyzing the environmental psychology aspect

This section visualizes data coming from Work Package 1 regarding the environmental psychology aspect. The data is provided in teams group, named WP1-Ålesund data_Day-Head torches.xlsx. The original data is in the excel format, as captured in Figure 20 below. The answers were scaled from 1 to 5 and represented on the **Mean** column. **SD** stands for standard deviation.

	Day light		Head torch regular		Head torch customized	
Variable	Mean	SD	Mean	SD	Mean	SD
Visual accessibility	4.91	.19	3.80	1.01	3.75	1.08
Prospect	4.04	1.51	3.27	1.19	3.26	1.32
Escape	3.44	1.23	3.04	1.15	2.83	1.30
Perceived safety	4.72	.41	3.45	1.18	3.70	1.40
Perceived comfort quality	-	-	4.04	1.09	4.33	1.22
Perceived Strength quality	-	-	3.57	1.39	3.24	1.33
Restorative potential	4.46	.56	3.88	.81	3.90	.96
Behavioural intention (choose)	4.54	.76	3.96	1.17	3.91	1.13
Behavioural intention (avoid)	1.39	1.03	1.79	1.22	2.19	1.29
Colour of light	4.63	.61	3.47	1.23	3.35	.25
Light pollution	4.48	.76	2.18	1.10	1.61	.69

Figure 20: The original data from WP1 regarding the environmental psychology aspect

The goal of this survey is to assess pedestrian's satisfaction on three different conditions; (1) on the daylight, (2) using a regular head torch, and (3) using a customized head torch. The pedestrian's satisfaction was evaluated through several factors as listed on the **Variable** column on Figure 20. The dataset was gathered by surveying pedestrians as they walked along the selected path shown in Figure 21. Data collection took place along the section of the path marked in red and yellow. Most responses were gathered around the midpoint of the stretch, where pedestrians first walked the path and then answered the questions while being immersed (half way).

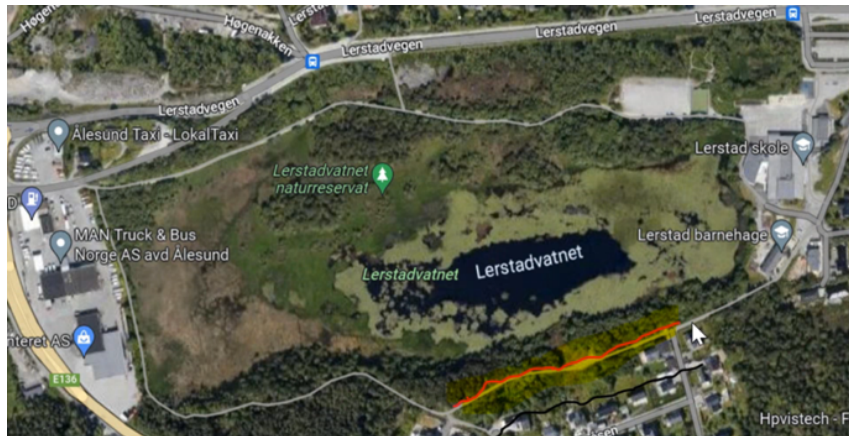


Figure 21: The selected path to access the environmental psychology aspect

Given that the Nordark digital twin only supports geojson format on the data visualization feature,

the original file needs to be converted to the geojson file. This new file is provided in teams group, named EnvironmentalPsychology.geojson. To meet the goal of this survey, we visualize the **Mean** of each condition and scale it between 0 to 1 in the heatmapped bar. To visualize the data on the digital twin, follow these steps:

- Go to *Data visualization* feature.
- Click the *Add dataset* button, and select the file "EnvironmentalPsychology.geojson"
- Zoom out/in and adjust the camera to find the analyzed path

To illustrate the comparison on three different conditions, we visualized all three conditions in a single frame, as shown in Figure 22, where 1, 2, and 3 correspond to daylight, the regular head torch, and the customized head torch, respectively.

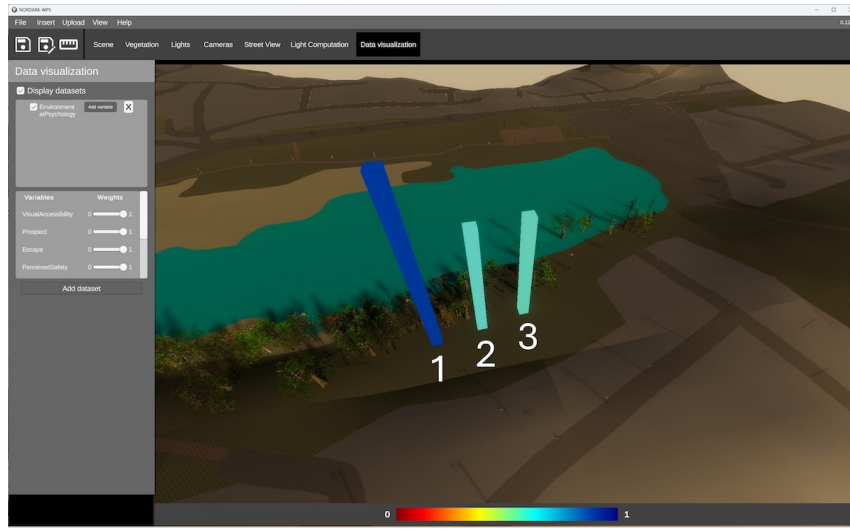


Figure 22: The analysis on the environmental psychology aspect, 1 for day light, 2 for regular head torch, 3 for customized head torch.

Figure 23 presents the results for different setups of the analyzed variables. 0 in red indicates lower satisfaction levels, while 1 in blue represents the highest satisfaction. Part (a) of Figure 23 considers only light pollution factor. It shows that the pedestrians have lower satisfaction on the light pollution produced by the customized head torch than the one from the regular head torch. On part (b) and (c), the pedestrians agreed that the customized head torch perceives a higher level of safety and comfort than the regular head torch. As shown in parts (c) and (d), perceived comfort and strength qualities were not assessed under the daylight condition.

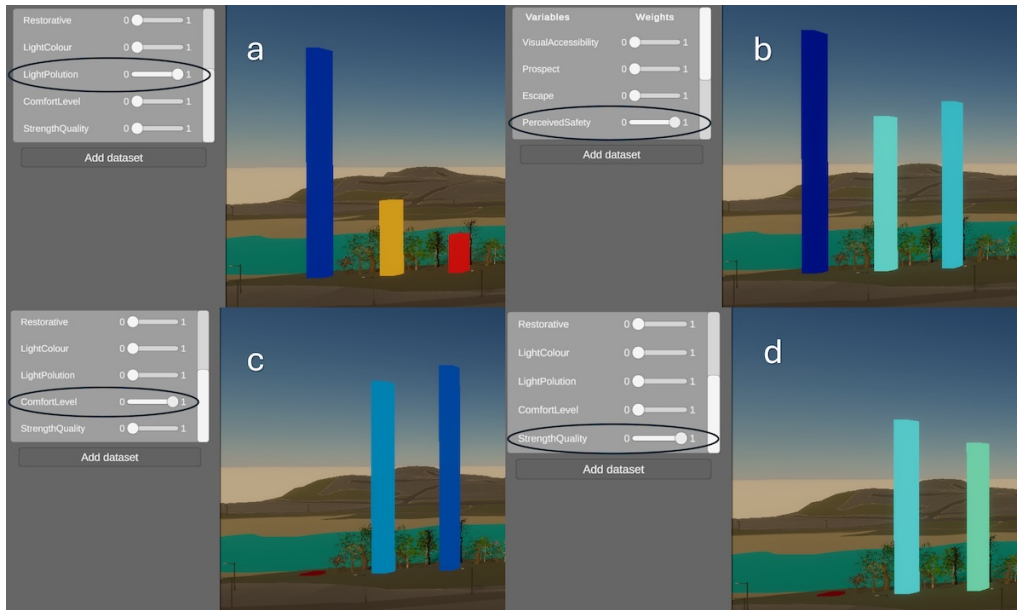


Figure 23: Data visualization on the environmental psychology aspect, (a) light pollution, (b) perceived safety, (c) perceived comfort level, (d) perceived strength quality.

5.2 WP4: Comparing the measured data with the simulated data

This section covers data coming from Work Package 4 regarding the light measurement in Savja, Uppsala. This data is provided in teams group, named `Light_Data_Uppsala_WP5.xlsx`. The light measurement was conducted on the location shown in Figure 24



Figure 24: The project location in Uppsala and chose light poles' positions for the field studies. Top: a capture from Google Maps. Bottom: a capture from the digital twin software

The area between pole number 2 and 3 in Figure 25 was selected as the main area of light measurements due to having a spacing of 30 metres, which represents the average spacing between poles on the walking route. By defaults, pole number 2 and 3 are identified as light 191 and 192, respectively, on the digital twin.

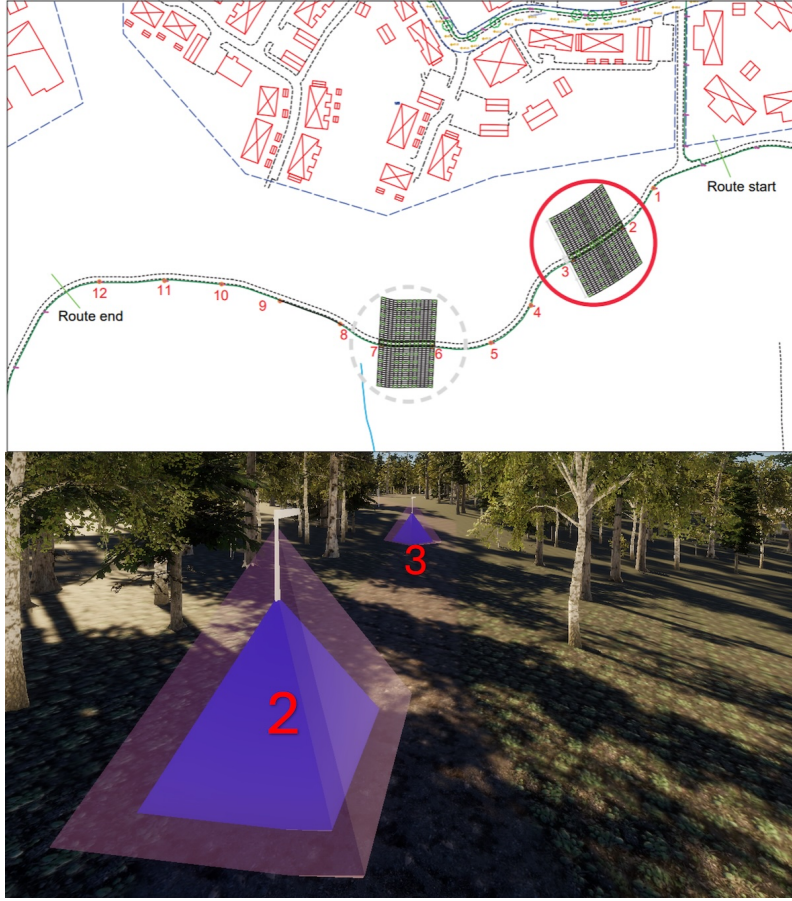


Figure 25: Area selected for the light measurements. Top: full detailed measurements between poles 2 and 3, and control measurements between poles 6 and 7. Bottom: a capture from the digital twin software, showing poles 2 and 3.

The grid for this light measurement is shown in Figure 26. Since the light pole is needed as a starting point for drawing lines, we can only visualize the measured data that can be drawn horizontally and vertically from the light pole, which are illustrated as blue boxes in Figure 26.

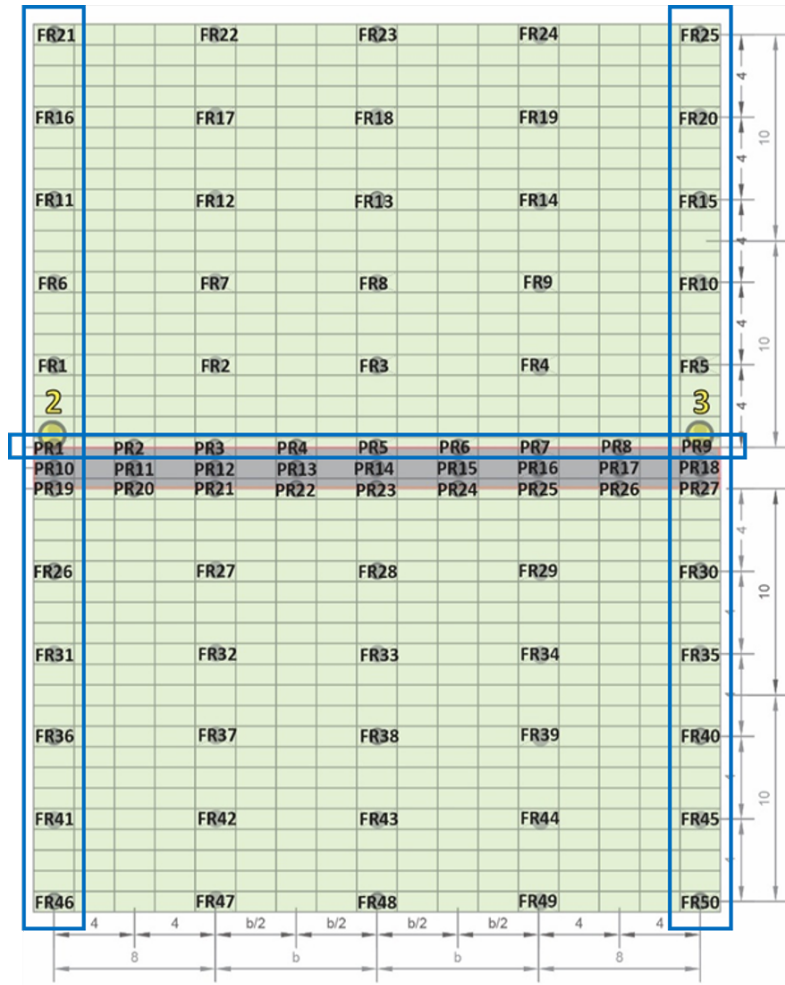


Figure 26: The light measurement grid for poles 2 and 3

The Nordark digital twin only supports geojson format to be imported, therefore, the original data needs to be converted to the geojson file. One converted file is provided as an example, named case3.geojson. This file includes the measurement between light poles: PR1, PR2, PR3, PR4, PR5, PR6, PR7, and PR8. The case is intervention light measurements, Scenario 1 100% (November - no snow) in Sävja, Uppsala - evening measurements - ground level (h: 0.00 m). To compare the measure data coming from WP4 and the modelled data on the digital twin, follow these steps:

1. Go to *Scene* feature, change the location to Uppsala.
2. Go to *Lights* feature, select the light poles 191 and 192, then change the IES file to *Prisma Light Elton 1-S 830 DNW ROTERAD*. Note: by default, the first light pole you see on the Uppsala scene is light pole 195, to find light poles 191 and 192, you need to move forward. If you are not sure, you can use Figures 24 and 25 to find the location of light poles 2 and 3 on the digital twin.
3. Go to *Scene* feature, set the date to 17 November 2023, the time to 18:30, the clouds to Overcast.
4. Go to *Light computation* feature, click *Create measure line around light pole* button, select "light 191", and set the parameter as shown in Figure 27, then click on *Create measure line*.

MEASURE LINE PARAMETERS X

Selected light pole: Light 191

Line length (in meters) 28

Line angle with the light pole (in degrees) 237

Distance between measure points (in meters) 4

Create measure line

Figure 27: The measure line parameters for PR1 to PR8

5. Go to *Light computation* feature, check the *Display graph* box to show the graph along the drawn line.
6. Click the *Import results* button, then select the case3.geojson file to import the light measurement data.
7. if you follow all stages properly, from setting the scene, date, time, clouds, to finding the light pole and creating the exact line, then you will get the results as shown in Figure 28.

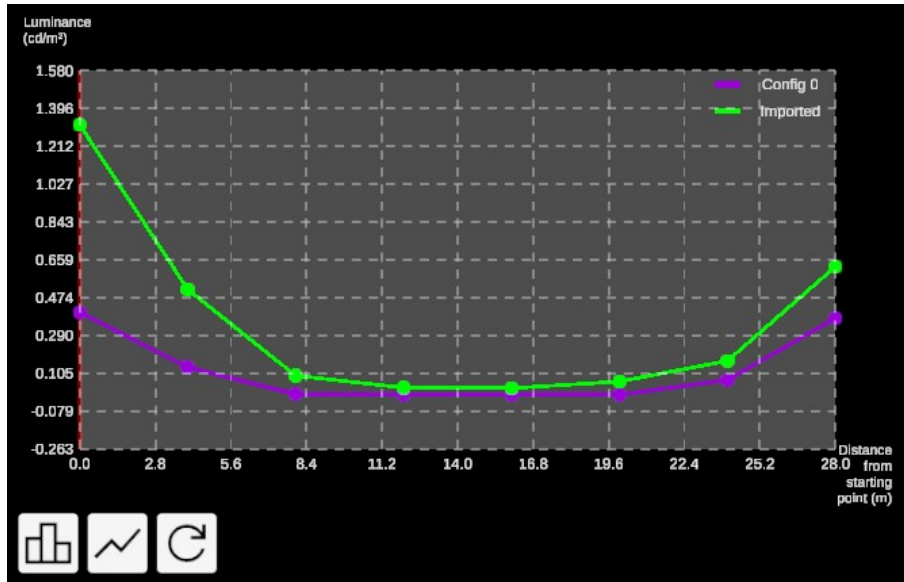


Figure 28: The comparison of the measured data and the modelled data on PR1 to PR8, at 18:30 on 17 November 2023, no snow. Purple line represents the modelled data, green line represents the measured data

The original light measurement data is provided in terms of illuminance (lux), whereas the digital twin system operates using luminance (lumen). To clarify the distinction between the two, as outlined in the *Technical Documentation – NORDARK DT*:

- Illuminance measures the amount of light being projected towards an object by the light (in lux).
- Luminance is the amount of light reflected off the surface being illuminated. It is considered the human perception of brightness.

In the digital twin, luminance is computed using color values and camera exposure. In a linear color space, the rendering-based formula is:

$$Luminance = (0.0396819152, 0.458021790, 0.00609653955) * pixelColor / exposure \quad (1)$$

For converting measured illuminance data (lux) to luminance (cd/m²), a common approximation is:

$$Luminance(cd/m^2) = Illuminance(lux) * \rho / \pi \quad (2)$$

where ρ is reflectance of the surface. In the digital twin, the surface between light poles 2 and 3 is identified as an asphalt path, with a reflectance value of 0.05–0.15. While the surface around the trees is identified as forest, with a reflectance value of 0.1–0.2. Therefore, a reflectance value will be changed corresponding to the area surface.

To create a measurement line for the points PR1, PR10, PR19, FR26, FR31, FR36, FR41, and FR46, use the line parameters shown in Figure 29(a). For the points PR1, FR1, FR6, FR11, FR16, and FR21, refer to the parameters in Figure 29(b). Both of these lines should be assigned to light 191.

To create a measurement line for PR9, PR18, PR27, FR30, FR35, FR40, FR45, and FR50, use the parameters from Figure 29(c). For the line consisting of PR9, FR5, FR10, FR15, and FR22, use the parameters in Figure 29(d). These two lines should be associated with light 192.

Panel	Selected light pole	Line length (m)	Line angle (deg)	Distance (m)
a	Light 191	22	327	4
b	Light 191	20	147	4
c	Light 192	22	327	4
d	Light 192	20	147	4

Figure 29: The measure line parameters for each line shown in Figure 26

Another example was made for measurement points of PR9, FR5, FR10, FR15, FR20, and FR25, with different scenario. That is intervention light measurements, Scenario 1 100% (January - snow) in Sävja, Uppsala - evening measurements - ground level (h: 0.00 m). The Geojson file is provided named case55.geojson.

1. Go to *Scene* feature, change the location to Uppsala.
2. Go to *Light computation* feature, click *Create measure line around light pole* button, select "light 192", and set the parameter as shown in Figure 29(d), then click on *Create measure line*.
3. Go to *Scene* feature, set the date to 13 January 2025, the time to 18:40, the clouds to Overcast, 35% snow.
4. Go to *Light computation* feature, check the *Display graph* box to show the graph along the drawn line.
5. Click the *Import results* button, then select the case55.geojson file to import the light measurement data.
6. if you follow all stages properly, you will get the results as shown in Figure 30.

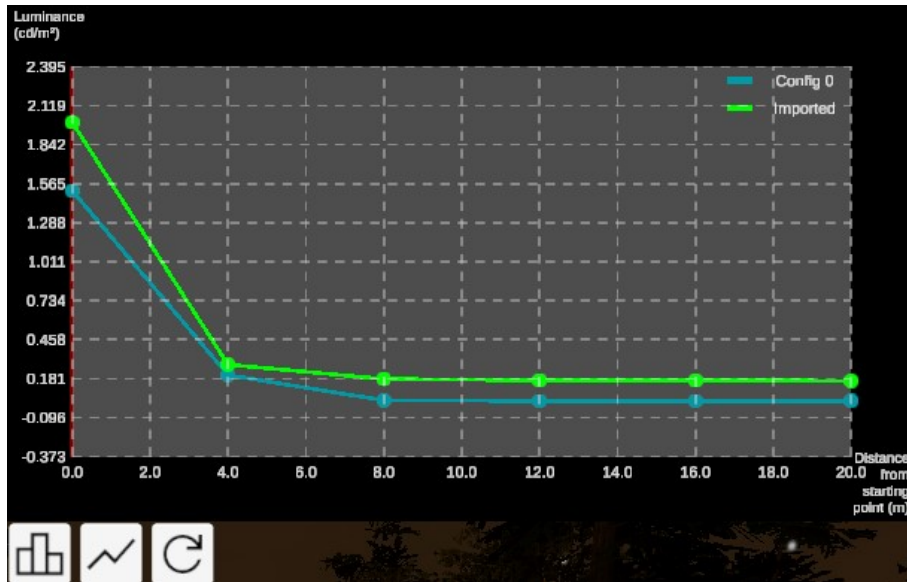


Figure 30: The comparison of the measured data and the modelled data on PR9, FR5, FR10, FR15, FR20, and FR25, on 13 January 2025, the time to 18:40, the clouds to Overcast, 35% snow. Blue line represents the modelled data, green line represents the measured data

5.3 WP6: Survey on green space and wildlife in the neighbourhood

This section presents data from Work Package 6, focusing on residents' opinions about their neighbourhood. The information was gathered through a questionnaire answered by local residents, addressing topics such as green spaces, wildlife, and vegetation in the area. Responses were categorized into five levels: strongly agree, agree, neutral, disagree, and strongly disagree. The questions are listed below:

- There should be more green space in my neighbourhood
- The green space in my neighbourhood is very important to me.
- I know a lot about the wild plants in my neighbourhood.
- I know a lot about the wild animals in my neighbourhood.
- Urban animals and plants have as much right as humans to exist.

To visualize the questionnaire data, two display options are available: (i) using the five responses as variables, or (ii) using the five questions as variables.

The first option is useful for identifying which questions received the most "strongly agree," "neutral," or "strongly disagree" responses. To visualize this data, use the provided GeoJSON file named `Arne_questionnaire_opinion.geojson` and follow these steps within the software:

- Go to *Data visualization* feature.
- Click the *Add dataset* button, and select the file "Arne_questionnaire_opinion.geojson"
- Zoom out/in and adjust the camera to find the analyzed path

Figure 31 illustrates the results, where the listed variables are strongly agree, agree, neutral, disagree, and strongly disagree. In this context, (1) represents the opinion that "there should be more green space in my neighbourhood." (2) corresponds to the opinion that "the green space in my neighbourhood is very important to me." (3) reflects the opinion that "I know a lot about the

wild plants in my neighbourhood.” (4) relates to the opinion that ”I know a lot about the wild animals in my neighbourhood.” (5) expresses the opinion that ”urban animals and plants have as much right as humans to exist.” Part (b) of Figure 31 indicates that local residents are most likely to feel neutral towards the idea of making the neighbourhood greener and their knowledge of wild plants. Part (c) of Figure 31 shows that residents strongly agree on the importance of green space and equal rights for all living beings. Part (d) of Figure 31 reveals that only a small number of respondents strongly disagree with any of the questions.

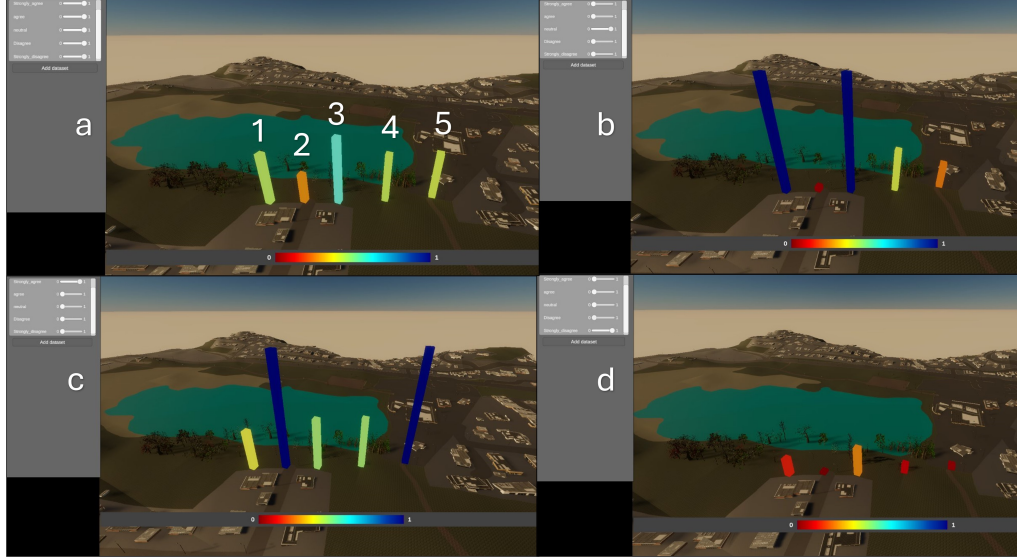


Figure 31: The first option to visualize the questionnaire information, where the five answers as the listed variables. (a) all answers are considered. (b) only the neutral answer is considered. (c) only the strongly agree answer is considered. (d) only the strongly disagree answer is considered.

The second option is useful for viewing residents’ responses to each individual question. To visualize this data, use the GeoJSON file named `Arne_questionnaire_5var.geojson` and follow these steps in the software:

- Go to *Data visualization* feature.
- Click the *Add dataset* button, and select the file ”`Arne_questionnaire_5var.geojson`”
- Zoom out/in and adjust the camera to find the analyzed path

Figure 32 displays the results, where the listed variables correspond to the numbers (1) through (5) from Figure 31, and are shortened as `greener`, `green_importance`, `wild_plants`, `wild_animals`, and `equal_right`. In Figure 32, (1) represents the answer ”strong agree”, (2) represents ”agree”, (3) represents ”neutral”, (4) represents ”disagree”, and (5) represents ”strong disagree”. Part (a) of Figure 32 indicates that residents generally agree with all the questions. Part (b) highlights that residents tend to feel neutral about the idea of making the area greener. Part (f) shows that residents strongly agree with the concept of equal rights among urban animals, plants, and humans.

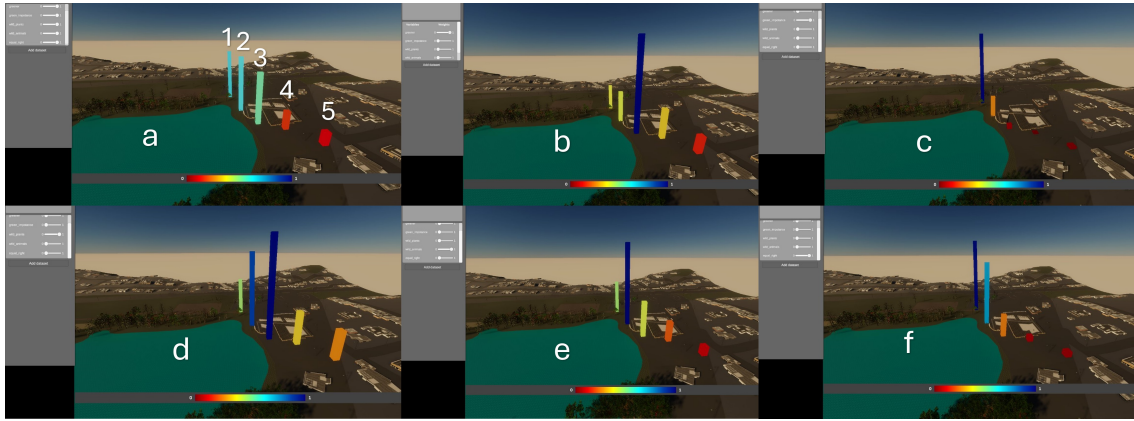


Figure 32: The second option to visualize the questionnaire information, where the five questions as the listed variables. (a) all questions are considered. (b) only the greener question is considered. (c) only the question regarding the importance of green space is considered. (d) only the question regarding the knowledge of wild plants is considered. (e) only the question regarding the knowledge of wild animals is considered. (f) only the question regarding the equal right over all living things is considered.

5.4 WP6: Urban lifestyle vs wildlife protection

On the questionnaire, an additional question was included to explore residents' views on the balance between a comfortable urban lifestyle and wildlife protection. The question was stated as: **How do you think about wildlife protection and urban lifestyle?** with the following response options:

- To protect urban wildlife and plants I have to accept restrictions in my lifestyle
- Protecting urban wildlife and plants is important, but not if it means changing my lifestyle
- To enjoy a pleasant and comfortable lifestyle, I can't avoid losing urban wildlife and plants

To visualize this data using the data visualization feature, a GeoJSON file named `lifestyle_wildlife.geojson` is provided. Follow these steps in the software:

- Go to *Data visualization* feature.
- Click the *Add dataset* button, and select the file "lifestyle_wildlife.geojson"
- Zoom out/in and adjust the camera to find the analyzed path

Figure 33 illustrates local residents' responses regarding their views on wildlife protection and urban lifestyle. The variables were defined based on the available response options. **wildlife&plants** refers to residents who prioritize urban wildlife and plants over their own lifestyle, expressing a willingness to accept lifestyle restrictions to support conservation. **both** represents those who believe protecting urban wildlife and plants is important—but not at the cost of changing their lifestyle. **lifestyle** reflects residents who prioritize a pleasant and comfortable lifestyle, even if it results in the loss of urban wildlife and plants.

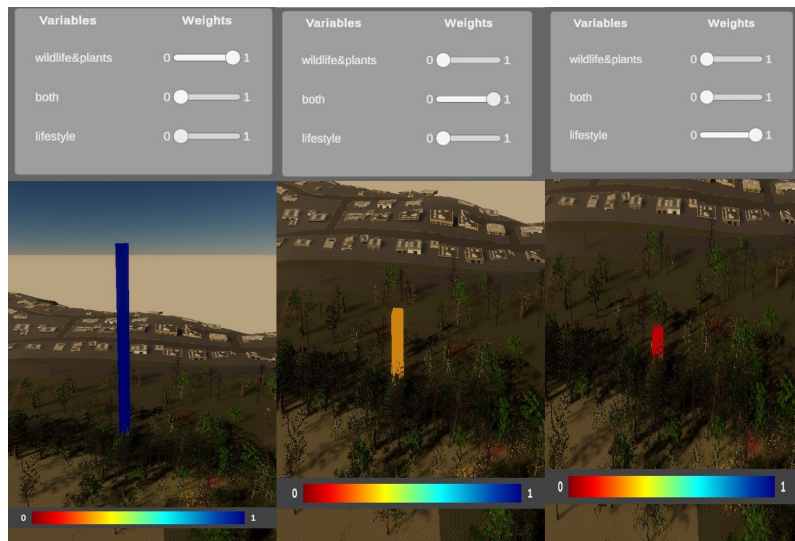


Figure 33: The questionnaire answer of how local residents think about wildlife protection and urban lifestyle.

6 Frequently Asked Questions

The authors believe that all necessary information for using and exploring the Nordark digital twin has been provided. However, this section is included to address users' frequently asked questions and offer additional guidance where needed.

- **What specifications my PC need to be able to run the digital twin software?**

It is a bit difficult to specify this, however we described the specification of our PC used for running the software in section 2. So if you can find something close (focus mostly on graphics card and memory) it should be fine.

- **How can I add more light poles on the selected scene?**

The answer is described in Section 3.3.

- **Can I create my own vegetation scenario? And how?**

Yes, the method is explained in Section 3.2

- **Can I simulate as if a human (child or adult) or an insect walks on the digital twin?**

Yes, *Street view* feature aims to achieve this simulation, as described in Section 3.5

- **How can I compare my measured data with the simulated data on the digital twin?**

For this purpose, you will use the *Light computation* feature. Start by preparing a geojson file containing your measured data. Next, set the digital twin to match the conditions during your measurement, such as the location, date, time, snow and cloud conditions, and the specific IES file of the light pole. After that, create a line around the light pole, display the graph, and then import your measured data file. Detailed examples and descriptions are provided in Section 5.2.

- **What kind of data I can import to the digital twin? And for what purposes?**

Only data in GeoJSON format can be imported into the system. There are two main features available for working with imported data: *Light computation* and *Data Visualization*.

Light computation feature is designed for comparing luminance data. This can serve various purposes: (i) comparing real-world light measurements with those in the digital twin, as described in 5.2, (ii) evaluating luminance data from an original versus a proposed design, as describe in 4.2, (iii) comparing different types of IES files and/or lighting poles, and so on.

Data Visualization feature allows you to display data that is quantitatively scalable. It represents values on a scale from 0 (indicating the lowest value) to 1 (indicating the highest value). It is suitable only for numerical data that can be meaningfully scaled within this range.

- **How can I convert my data from excel format to geojson format?**

It depends on your purpose. If you want to convert your data to geojson for *Light Computation* purpose, the full answer to this question is explained in Section 4.3. It covers how to find the coordinates, provides an example CSV file, and includes a Python script to help with the conversion process.

If you want to convert your data for *Data Visualization* purpose, the answer to this question is described in Section 4.4 including the example case.

- **What the data visualization feature can and can not do?**

First of all, the data visualization feature only supports files in the GeoJSON format. This format requires a geometry property that specifies the location on the data. The geometry can be either a Point or a LineString, both of which use latitude and longitude coordinates. You can obtain these coordinates from Google Maps or by following the instructions provided in Section 4.3.

Secondly, the data visualization feature displays information using a scale from 0 to 1 in both the heatmap and bar chart formats. For example, as outlined in Section 3.7, the walkability score ranges from 0 (red), indicating areas that are difficult to walk in, to 1 (blue), indicating areas that are easy to walk in, based on selected variables. Similarly, in Section 5.1, the satisfaction metric follows the same scale: 0 (red) signifies the least satisfied, while 1 (blue) indicates the most satisfied, evaluated under three different conditions using relevant variables.

If the data you wish to visualize cannot be represented on a scale from 0 to 1, it cannot be displayed using this digital twin’s data visualization feature. For instance, the dataset from Work Package 6, which explores locations from the local species perspective, does not fit this format. In that project, youth participants imagined themselves as specific animals (e.g., frogs, dogs, ducks) and described what those species might feel while moving through a given area. These responses consist of qualitative descriptions that cannot be reduced to a numerical scale. The same limitation applies to the dataset related to the VR prototype from Work Package 6. However, such data can still be explored using *Screen View* feature in the future.

7 Potential Future Direction

In this section, we discuss about potential implementations of the digital twin in the future, based on the available data from other Work Packages.

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Appendix